

Topics:

- Frequency combs
- Heterodyne detection
- Lineshapes

Resources:

- All things optics: [RP Photonics Encyclopedia](#)
- Lineshapes and molecular spectroscopy: [HITRAN Database](#)
- [Time and Frequency from A to Z](#)
- Textbooks in Ladd

Relevant Papers (work in progress):

Frequency Combs:

1. [20 years of developments in optical frequency comb technology and applications](#)
2. [Optical frequency combs: Coherently uniting the electromagnetic spectrum](#)
3. [Frequency comb spectroscopy](#)
4. [Femtosecond optical frequency combs: Principle, Operation and Applications](#)
5. <https://www.nist.gov/topics/physics/optical-frequency-combs>

Laser Heterodyne Radiometry (LHR):

6. [Frequency Comb Calibrated Laser Heterodyne Radiometry for Precision Radial Velocity Measurements](#)
7. [Thermal-light heterodyne spectroscopy with frequency comb calibration](#)
8. [Precision Doppler shift measurements with a frequency comb calibrated laser heterodyne radiometer](#)
9. [Thermal infrared laser heterodyne spectroradiometry for solar occultation atmospheric CO₂ measurements](#)
10. [Atmospheric trace gas measurements using laser heterodyne spectroscopy](#)
 - a. In “PDFs folder”
 - b. Most useful through Section 4.2.1

Radial Velocity Measurements and Stellar Variability

11. [Extreme Precision Radial Velocity Working Group Final Report](#)
12. [State of the Field: Extreme Precision Radial Velocities](#)
13. [The Extreme Stellar-signals Project. III. Combining Solar Data from HARPS, HARPS-N, EXPRES, and NEID](#)

Laser-Based Sensing:

14. [Spectroscopy and Optical Diagnostics for Gases](#)
 - a. Chapters 1, 2, 4, 7 and 8
15. [Baseline-free quantitative absorption spectroscopy based on cepstral analysis](#)

Absorption Line Shapes and Molecular Spectroscopy:

16. [The effects of advanced spectral line shapes on atmospheric carbon dioxide retrievals](#)
17. [The air-broadened, near-infrared CO₂ line shape in the spectrally isolated regime: Evidence of simultaneous Dicke narrowing and speed dependence](#)
18. [Spectroscopy and Optical Diagnostics for Gases](#)
19. [High-temperature absorption line shape parameters for CO₂ in the 6800–7000 cm⁻¹ region from dual frequency comb measurements up to 1000 K](#)
20. [Measurement and Modeling of Air-Broadened Methane Absorption in the MERLIN Spectral Region at Low Temperatures](#)
21. [Precise methane absorption measurements in the 1.64 μm spectral region for the MERLIN mission](#)
22. [Recommended isolated-line profile for representing high-resolution spectroscopic transitions \(IUPAC Technical Report\)](#)
23. [HITRAN Database](#) and associated [definitions](#).
24. [HITRAN Application Programming Interface \(HAPI\)](#)
25. Spectra of Atoms and Molecules (textbook by Bernath, likely in Ladd)